

Que. Write the full form and one-line use of: HTTP, HTTPS, FTP, and SMTP.

Ans.

➤ **HTTP – Hypertext Transfer Protocol**

Used to transfer web pages and data between a web browser and a web server.

➤ **HTTPS – Hypertext Transfer Protocol Secure**

Used to securely transfer web pages and data over the internet using encryption.

➤ **FTP – File Transfer Protocol**

Used to upload and download files between a computer and a server.

➤ **SMTP – Simple Mail Transfer Protocol**

Used to send emails from one mail server to another.

Que. What is the difference between IPv4 and IPv6? Answer in 2 lines.

Ans.

- **IPv4** uses a **32-bit** address (e.g., 192.168.1.1), while **IPv6** uses a **128-bit** address (e.g., 2001:db8::1).
- **IPv6** supports many more unique IP addresses and provides better security and efficiency than **IPv4**.

Que. Run nslookup google.com on your system and write down the IP address you get.

Ans.

```
Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\WHO ARE YOU>nslookup google.com
Server: UnKnown
Address: 10.57.200.238

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4000:1010::8a
           2404:6800:4000:1010::8b
           2404:6800:4000:1010::71
           2404:6800:4000:1010::65
           142.250.70.46
```

Que. What do these HTTP status codes mean: 200, 301, 404, 500, and 403?

Ans.

- **200 OK** – The request was successful, and the server returned the requested resource.
- **301 Moved Permanently** – The requested page has been permanently moved to a new URL.
- **404 Not Found** – The requested page or resource could not be found on the server.
- **500 Internal Server Error** – The server encountered an unexpected error while processing the request.
- **403 Forbidden** – The server understands the request but denies permission to access the resource.

Que. Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

Ans.

- **Root (.)** – The highest level of the DNS hierarchy that directs requests to the correct Top-Level Domain (TLD) server.
- **TLD (Top-Level Domain): .com** – Identifies the top-level domain and points to the servers responsible for **google.com**.
- **Domain: google** – The registered website name owned by Google.
- **Subdomain: www** – A subdomain used to identify the web service of the **google.com** domain.

Example:

```
Root (.)
└── .com (TLD)
    └── google (Domain)
        └── www (Subdomain)
```